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Assesment Tool-3:

1 A smart home automation system is being designed to manage lighting, temperature and security based on user behaviour and environmental conditions. The system must respond to real time inputs such as motion detection, temperature changes and time of day while also learning user preferences over time. In this context, explain the different types of itelligent agents

A Intelligent agents in a smart home:

Simple-reflex: uses condition action rules on the current percept only.

Example: if motion detected and it is night, turn on the light. It is fast and cheap but has no memory, so it can't tell whether a room is empty or the user is just sitting still

model based reflex: keeps an internal state of home. E

Example:- it refers the user is still in the living room depite no motion for 5 minutes, so it keeps the lights on

Goal based reflex: Chooses actions to reach a goal such as "room at 24°C by 7PM" or house secured by 11pm. It plans ahead for example precooling the house before user returns it is flexible but does not weigh trade-offs

Utility-based: Maximizes a utility function that balances confort, energy cost and Securily

Example: when comfort and electricity cost conflict, it Picks the AC setting with the best overall score, It also copes with uncertainty

Best choice: a hybrid, with a learning component
- Reflex rules for immediate safety and security, where speed matter
- A model based layer for tradeck the home's state
- Goal based planning for scheduled routines
- A utility-based layer for trade-offs between comfort and energy
- A learing element that updates user perferences over time

This combination gives fast reactions context awareness and efficiency without sacrificing comfort

2 A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like uniform cost search and Iterative Deepening search (IDS) can help solve this problem. Discuss the advantages and disadvantages of those algorithms in terms of intelligent agents. and expain how agent design affects decision making

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Uniform Cost Search (UCS):

Expands the node with the lowest path cost (g(n)) using a priority queue. In a maze with varying steps costs, it finds the cheapest path. It is complete and optimal if costs are positive

Iterative Deepening Search (IDS): R

Runs depth-limited DFS with limits 0, 1, 2 and soon. It is complete, optimal for uniform step costs and uses little memory. It suits an unknown maze because the goal depth is not known

UES pros and cons: Handles variable costs, but its memory use is exponential and it explores in all directions

O IDS pros and cons: Very memory-efficient, but it re-expands upper levels (a small over head, since most nodes are at the deepest level) and ignores step costs

Link to agent types:

O A simple reflex agent (e.g. "turn left if wall ahead") can loop forever in a maze

- A model based agent remembers visited cells and builds a map
- A goal based agent uses search to plan a path to the exist
- A utility-based agent chooses among paths using time, energy and risk

Agent design decides what the robot remembers and how it chooses between actions, so the search strategy only works well if the agent has a model of the maze

Code of Cadnduct

I A. Charith Reddy/ 1965960111 certify that this submission is my original work and that I have adhered to the guide lines specified for this assesment. I understand that any violation of academic integrity rules will result in disciplanary action.

Signature of student

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